

JavaScript Mastery • Live Coding Session

Day 9 — String Methods Deep Dive

Mastering text manipulation, validation, search queries, immutability, and hands-on debugging.

Duration: ~30 Minutes | Camera On • Interactive Practice

Today's Covered Methods & Topics

1. Core Fundamentals

- What is a String & Immutability recap
- Why String Methods Matter (search, validation, casing, trim)
- length, charAt(), indexOf()

2. Portion Extraction & Casing

- slice() vs substring() (negative index rule)
- toUpperCase() & toLowerCase()

3. Cleaning, Search & Replacement

- trim(), trimStart(), trimEnd()
- includes(), startsWith(), endsWith()
- replace() vs replaceAll()

4. Array Combo & Challenge

- Bonus: split() and join()
- Side-by-side comparison matrix
- Complete rawInput 5-step Homework Challenge

CONCEPT RECAP

What is a String & Immutability Rule

- String = text data inside quotes (" ", ' ', or template ` `)
- Strings in JS are **immutable** (cannot be directly mutated in-place)
- Methods always return a **brand new string**, never touch original

```
let name = "Ahmed"; let city = 'Karachi'; let sentence = `I live in ${city}`; // Template literal let text =
```

💡 **Key Takeaway:** Jitne bhi methods seekhoge, sab new string return karte hain!

Why String Methods Matter



Search Bar

User input ko lowercase karke match karna case-insensitive search ke liye.



Form Validation

Email ke andar '@' check karna ya domain verify karna.



Name Formatting

Pehla letter capital karna (jaise 'ahmed' ko 'Ahmed' banana).



Trimming Extra Spaces

Form inputs ke aage/peeche accidental spaces hatana.

METHOD GROUP 1

length, charAt(), indexOf()

- **length**: property (not a function), returns total length
- **charAt(index)**: returns character at specific index
- **indexOf(val)**: returns first matching position, or **-1** if not found
-  **-1 check** is heavily used for conditional existence logic

length, charAt(), indexOf() in Action

```
let name = "JavaScript"; console.log(name.length); // 10 console.log(name.charAt(0)); // "J" console.log(name.indexOf("A"));
```

Mini Practice: indexOf()

 Try it yourself

Batao is string mein '**Script**' kahan se start hota hai?

Try: `name.indexOf('Script')`

[Pause video](#) / [Live try in console...](#)

slice() vs substring() (Portion Extraction)

- Both extract a substring portion from a string
- **slice(start, end)**: supports negative indices (counts from end)
- **substring(start, end)**: treats negative numbers as 0
- 🔥 Pro developers prefer **slice()** due to negative index flexibility

slice() vs substring() Code Demo

```
let text = "Hello World"; console.log(text.slice(0, 5)); // "Hello" console.log(text.substring(0, 5)); // "H"
```

Negative Index Behavior Comparison

✓ `slice(-5)`

Counts backwards 5 steps from end of string.

```
text.slice(-5) // "World"
```

✗ `substring(-5)`

Clamps negative value to index 0.

```
text.substring(-5) // "Hello World"
```

🧠 Mini Practice

`text.slice(6)` likh kar dekho — kya result aata hai?

Answer: 'World', kyun ke end value nahi di toh end tak chala jata hai

toUpperCase() & toLowerCase() + Search Match

- Simple case conversion methods
- Essential for case-insensitive search queries

```
let searchQuery = "AHMED"; let userName = "ahmed khan"; if (userName.toLowerCase().includes(searchQuery.toLo
```

METHOD GROUP 4

trim(), trimStart(), trimEnd()

- Removes surrounding whitespace characters
- Critical for database cleanliness in form inputs

```
let messyInput = " Hello World "; console.log(messyInput.trim()); // "Hello World" console.log(messyInput.tr
```

Mini Practice

Ek variable banao `let username = ' Bilal '`; aur usko trim karke uska length print karo.

METHOD GROUP 5

includes(), startsWith(), endsWith()

- Return boolean (`true` / `false`)
- Perfect foundation checks for validation

```
let email = "student@gmail.com"; console.log(email.includes("@")); // true console.log(email.startsWith("stu
```

Simple Email Validation Function

```
function isValidEmail(email) { if (email.includes("@") && email.endsWith(".com")) { return true; } return false; }
```

Mini Practice

Check karo ke ek phone number string '0300-1234567' '0300' se start hoti hai ya nahi, startsWith() use karke.

replace() vs replaceAll() Difference

replace()

Sirf **pehla match** replace karta hai.

```
sentence.replace("JS", "React") // 1st match only
```

replaceAll()

Saari **matching occurrences** replace karta hai.

```
sentence.replaceAll("JS", "React") // All occurrences
```

⚠ Common beginner confusion — dhyan se practice karna!

replace() / replaceAll() & Phone Cleaning

```
let sentence = "I love JavaScript. JavaScript is amazing."; console.log(sentence.replace("JavaScript", "React"))
```

Real-world example — Removing unwanted characters:

```
let phoneNumber = "0300-123-4567"; let cleanNumber = phoneNumber.replaceAll("-", ""); console.log(cleanNumber)
```

BONUS COMBO

split() and join() Powerful Combo

- **split(separator)**: converts string into an array
- **join(separator)**: converts array back to a string

```
let fullName = "Ahmed Ali Khan"; let nameParts = fullName.split(" "); console.log(nameParts); // ["Ahmed",
```

String Methods Summary Table

Method	What it does	Returns
charAt(i)	Specific index ka character deta hai	Single character
indexOf(x)	Character/word ki position batata hai	Number (-1 agar nahi mila)
slice(start, end)	String ka hissa nikalta hai (negative index bhi chalta hai)	New string
toUpperCase() / toLowerCase()	Case badalta hai	New string
trim()	Aage peeche ki spaces hatata hai	New string
includes()	Kuch string ke andar hai ya nahi	true/false
startsWith() / endsWith()	Shuru/akhir check karta hai	true/false
replace() / replaceAll()	Text replace karta hai (1 ya sab)	New string
split()	String ko array mein todta hai	Array
join()	Array ko string mein jorta hai	String



Homework / Assignment

```
let rawInput = " ahmed.ali@GMAIL.com ";
```

1. Pehle spaces trim karo (`trim()` use karke).
2. Poora string lowercase mein convert karo (`toLowerCase()`).
3. Check karo ke yeh valid email hai ya nahi — `includes('@')` aur `endsWith('.com')` use karke, aur result console pe print karo.
4. `split('@')` use karke email ko username aur domain mein alag karo, aur dono ko separately print karo.
5. Username ka pehla letter capital karo (hint: `charAt(0)` aur `slice(1)` combine karo) — aur final result print karo.

💬 Comments section mein apna solution zaroor share karna — winner mentioned next session!

SESSION WRAP-UP

Quick Recap & Key Takeaways

-  Strings immutable hoti hain — har method naya string return karta hai
-  slice() negative index samajhta hai, substring() nahi
-  includes(), startsWith(), endsWith() — validation ke liye perfect
-  replace() sirf pehla match badalta hai, replaceAll() sab kuch
-  split() aur join() — string aur array ke beech convert karne ka tareeka



Agla Session Preview

Scope (global/block/function), Hoisting, aur Closures ka intro — advanced & exciting topic!

Allah Hafiz, aur keep coding! 